

2 Nitrogen

Nitrogen drives canopy growth and protein formation. Its soil forms are highly affected by biology, water, oxygen, and temperature.

Why it matters

- **Growth and chlorophyll.** Nitrogen supports green leaf area, enzymes, proteins, and new shoots.
- **Balance matters.** Too little limits growth; too much can promote excessive vegetative growth, increase salt load, and raise loss to water or air.

Tentative leaf guide

Measure	Interpretation note
Tentative avocado range	UC guidance lists less than 1.6 percent as deficient, 1.8 to 2.4 percent as adequate, and more than 2.8 percent as excess.
Sampling caution	Season, leaf age, cultivar, crop load, water status, and laboratory method can shift the result.

Main soil and environmental controls

Factor	What it can mean in the root zone
pH and microbes	Microbes convert organic N and ammonium to plant-available forms. These processes slow when soil is cold, very dry, waterlogged, very acidic, or highly saline.
Organic matter	Residues can release N slowly. Fresh high-carbon material can temporarily tie up N while microbes decompose it.
Texture and leaching	Nitrate moves readily with water and is easily lost below roots in sandy soil or with long irrigation sets. Ammonium is held more strongly until converted.
Irrigation and oxygen	Water distributes N, but saturation can cause denitrification and weak roots. Uneven irrigation creates both leached and underfed zones.
Salinity and roots	Salt stress and root disease reduce uptake. Concentrated fertilizers can add local salt injury near shallow feeder roots.

Field clues and checks

- **Look for pattern and age.** General pale color and weak growth may suggest low N, but root rot, waterlogging, salinity, cold soil, and poor irrigation can look similar.
- **Check the full N budget.** Include fertilizer, irrigation-water nitrate, compost, manure, mulch breakdown, and expected crop demand.
- **Measure water movement.** Match small applications to root activity, soil storage, irrigation uniformity, and the risk of nitrate moving below roots.

DST interpretation A nitrogen flag should trigger a supply and loss budget plus a root and irrigation check. It should not automatically increase the fertilizer rate.